KENNITECH

Small Office VLAN
Network Report

Implementation, security configuration and verification

Project detail	Information
Project	Project 01: Small Office VLAN Network
Environment	Cisco Packet Tracer 9.x on macOS
Implementation date	7 November 2026
Network devices	Cisco 2911 router, Cisco 2960-24TT switch, six PCs
Departments	Sales, IT and HR
Addressing	IPv4, three /24 departmental subnets
Prepared for	KenniTech networking project portfolio

Project outcome
A segmented small-office network was built using three VLANs, router-on-a-stick inter-VLAN routing, DHCP address assignment and named extended IPv4 ACLs. The final endpoint tests match the required department access policy.

1. Executive summary

The network separates Sales, IT and HR into VLANs 10, 20 and 30. A Cisco 2960 switch provides Layer 2 connectivity. A Cisco 2911 router provides the default gateway for each VLAN through 802.1Q subinterfaces on a router-on-a-stick link. DHCP is configured on the router, and extended ACLs restrict traffic initiated by Sales and HR while permitting selected return traffic.

Evidence supplied on 7 November 2026 shows the topology, active VLANs, trunk status, router subinterfaces, DHCP leases, ACL entries, ACL interface attachment and final ping tests. The router and switch configurations were copied to startup configuration and Packet Tracer returned [OK].

2. Scenario, objectives and security policy

The scenario is a small office with three departments. Each department has two PCs and its own IPv4 subnet. The objective is to provide inter-VLAN connectivity where permitted while preventing unauthorised access between departments.

Source	Destination	Required policy
Sales	IT	Block Sales-initiated access
Sales	HR	Block Sales-initiated access
IT	Sales and HR	Allow IT-initiated access
HR	IT	Block HR-initiated access
HR	Sales	Allow HR-initiated access

Two inbound extended ACLs are applied on the Sales and HR router subinterfaces. No inbound ACL is attached to the IT subinterface in the supplied evidence. Return-traffic entries are placed before deny entries.

3. Topology and equipment

One Cisco 2911 router connects to one Cisco 2960-24TT switch. Six PCs connect to access ports, with two PCs per department. The router-to-switch uplink carries VLANs 10, 20 and 30 using IEEE 802.1Q tagging.

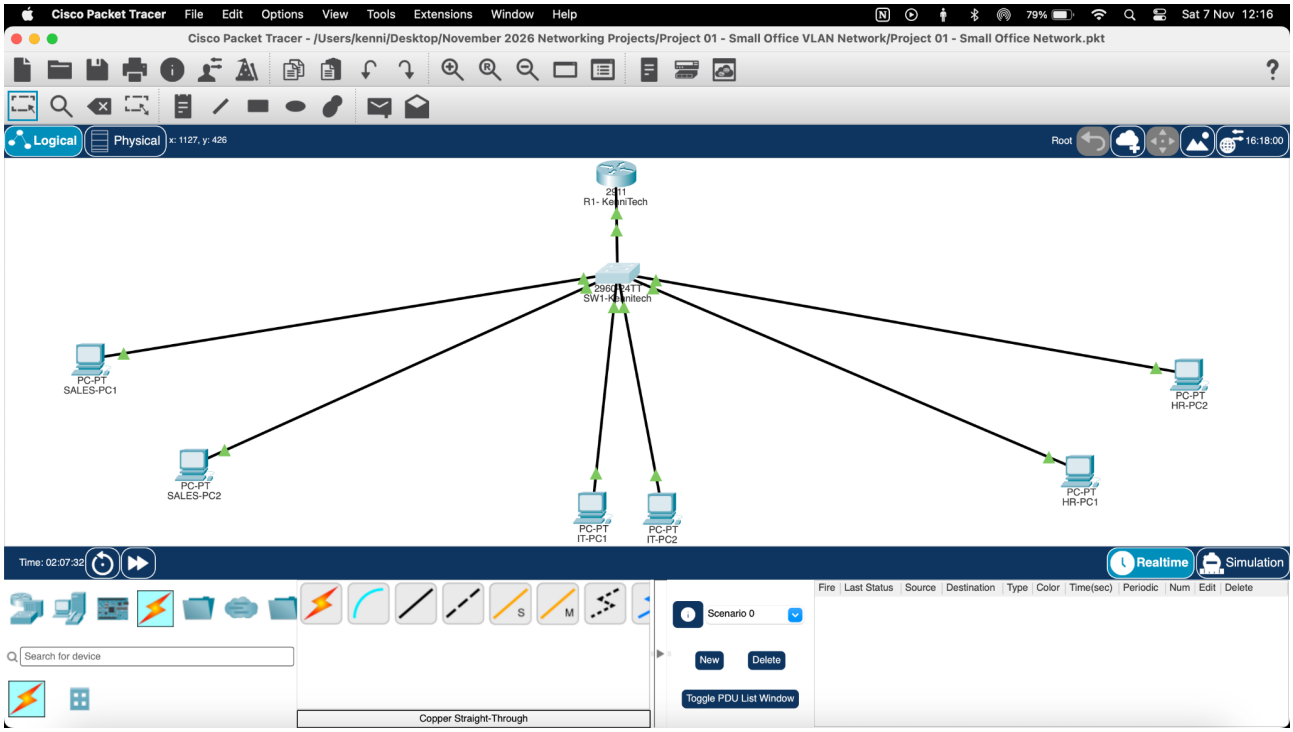


Figure 1. Packet Tracer logical topology showing R1, SW1 and six departmental PCs.

Device	Model / role	Purpose
R1-KenniTech	Cisco 2911	Inter-VLAN routing, DHCP and ACL enforcement
SW1-KenniTech	Cisco 2960-24TT	VLAN segmentation, access switching and trunk
Sales-PC1 and Sales-PC2	PC endpoints	Sales clients, VLAN 10

Device	Model / role	Purpose
IT-PC1 and IT-PC2	PC endpoints	IT clients, VLAN 20
HR-PC1 and HR-PC2	PC endpoints	HR clients, VLAN 30

4. VLAN and IPv4 addressing plan

Department	VLAN	Network / mask	Gateway	DHCP leases evidenced
Sales	10	192.168.10.0/24	192.168.10.1	192.168.10.11, .12
IT	20	192.168.20.0/24	192.168.20.1	192.168.20.11, .12
HR	30	192.168.30.0/24	192.168.30.1	192.168.30.11, .12

Subnet mask: 255.255.255.0 (/24). The router DHCP configuration excludes .1 through .10 in each subnet. The DHCP binding output shows two automatic leases in each departmental subnet.

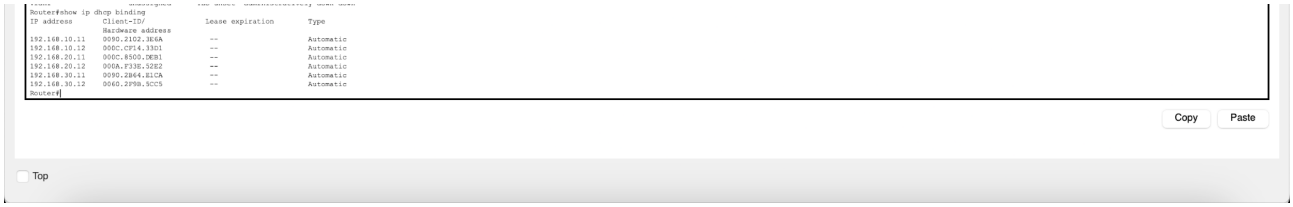


Figure 2. Switch VLAN membership output: Sales Fa0/1-Fa0/2, IT Fa0/3-Fa0/4 and HR Fa0/5-Fa0/6.



Figure 3. GigabitEthernet0/1 trunk verification, with VLANs 10, 20 and 30 active.

5. Switch configuration and verification

The switch access port allocation is Sales VLAN 10 on Fa0/1 and Fa0/2, IT VLAN 20 on Fa0/3 and Fa0/4, and HR VLAN 30 on Fa0/5 and Fa0/6. GigabitEthernet0/1 is an 802.1Q trunk. The output lists VLANs 1, 10, 20 and 30 as active and forwarding.

Verification and save commands:

```
show vlan brief<br/>show interfaces trunk<br/>copy running-config startup-config
```

The switch save operation completed and Packet Tracer displayed [OK].

6. Router configuration and verification

Subinterface	VLAN	Gateway	State
Gi0/0.10	10	192.168.10.1/24	up / up
Gi0/0.20	20	192.168.20.1/24	up / up
Gi0/0.30	30	192.168.30.1/24	up / up

The router uses 802.1Q subinterfaces for the three departmental VLANs. The show ip interface brief output reports all three as up/up. The DHCP binding table lists six client leases.

Verification and save commands:

```
show ip interface brief<br/>show ip dhcp binding<br/>show running-config<br/>copy running-config
startup-config
```

The router save operation completed and Packet Tracer displayed [OK].

7. Extended ACL design and placement

SALES-IN is applied inbound to GigabitEthernet0/0.10. HR-IN is applied inbound to GigabitEthernet0/0.30. The ACLs filter traffic as it enters the router from the relevant department.

7.1 SALES-IN

Seq.	Rule summary	Purpose
10	permit tcp Sales to IT established	Permit established TCP return traffic
20	permit icmp Sales to IT echo-reply	Permit replies to IT-originated ping
30	permit tcp Sales to HR established	Permit established TCP return traffic
40	permit icmp Sales to HR echo-reply	Permit replies to HR-originated ping
50	deny ip Sales to IT	Block other Sales-originated IP traffic
60	deny ip Sales to HR	Block other Sales-originated IP traffic
70	permit ip any any	Permit other traffic not matched earlier

7.2 HR-IN

Seq.	Rule summary	Purpose
10	permit tcp HR to IT established	Permit established TCP return traffic
20	permit icmp HR to IT echo-reply	Permit replies to IT-originated ping
30	deny ip HR to IT	Block other HR-originated IP traffic
40	permit ip any any	Permit other traffic, including HR to Sales

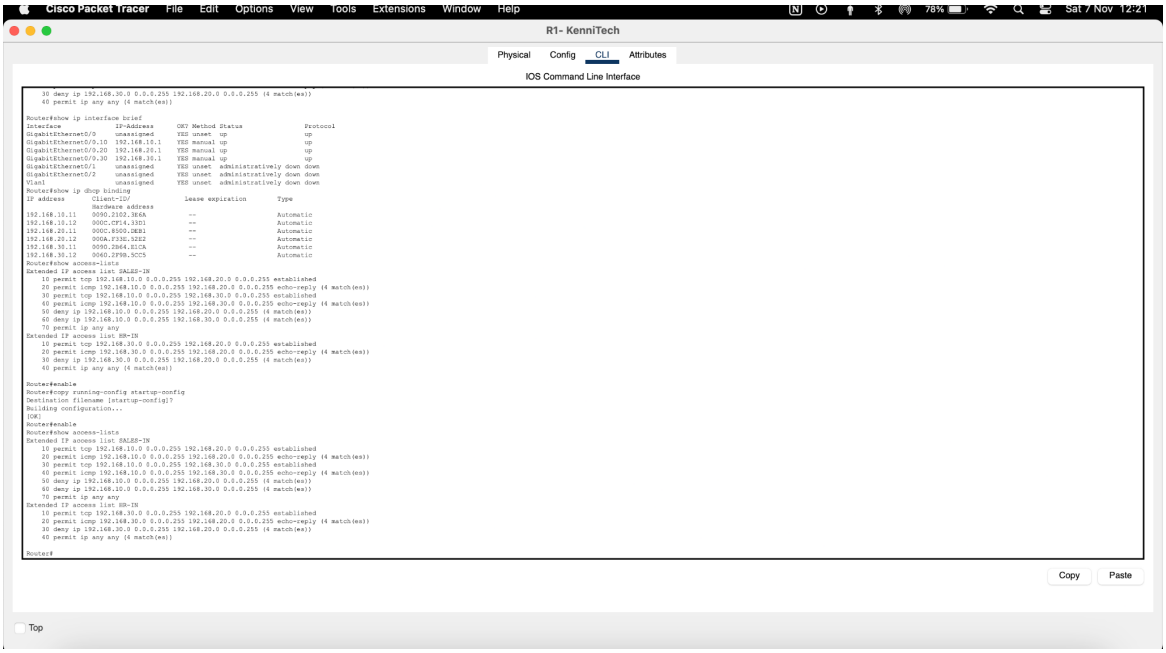


Figure 4. Router ACL entries and packet match counters.

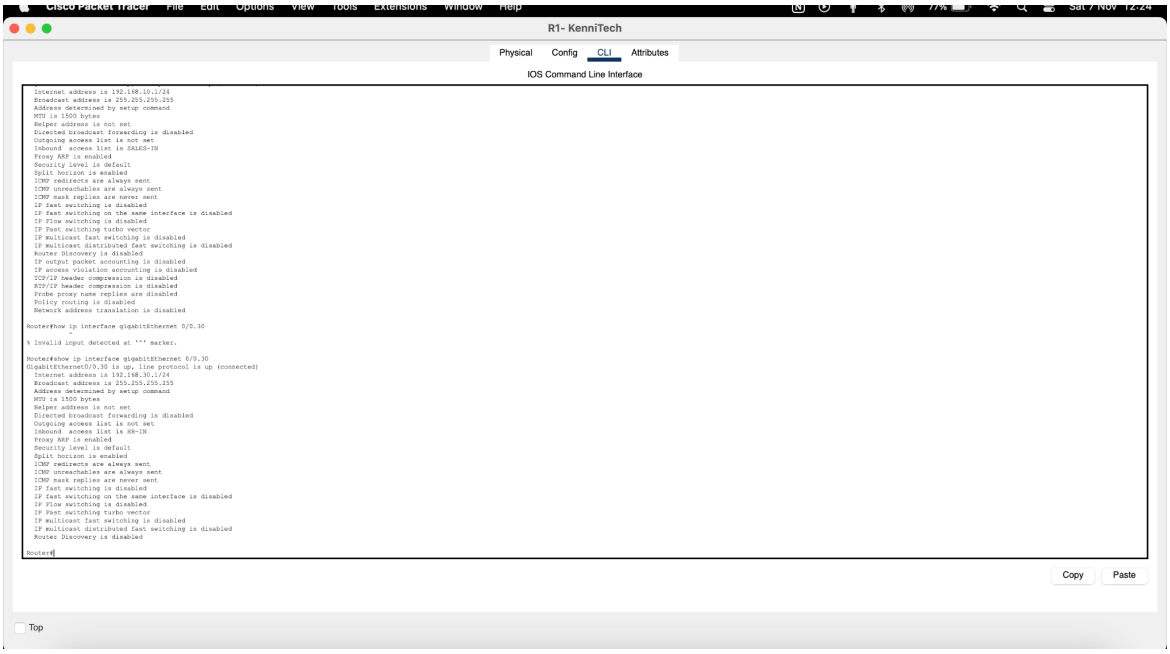


Figure 5. Interface output confirming SALES-IN and HR-IN are attached inbound.

8. Connectivity testing and results

The final command-prompt screenshots were checked against the required access policy. Sales tests show blocked access to IT and HR. IT tests show successful replies from Sales and HR. HR tests show blocked access to IT and successful access to Sales.

Source	Destination	Expected	Observed
Sales-PC1	IT-PC1 192.168.20.11	Blocked	Blocked, destination host unreachable
Sales-PC1	HR-PC1 192.168.30.11	Blocked	Blocked, destination host unreachable
IT-PC1	Sales-PC1 192.168.10.11	Allowed	4 replies, 0% loss
IT-PC1	HR-PC1 192.168.30.11	Allowed	4 replies, 0% loss
HR-PC1	IT-PC1 192.168.20.11	Blocked	Blocked, destination host unreachable
HR-PC1	Sales-PC1 192.168.10.11	Allowed	4 replies, 0% loss

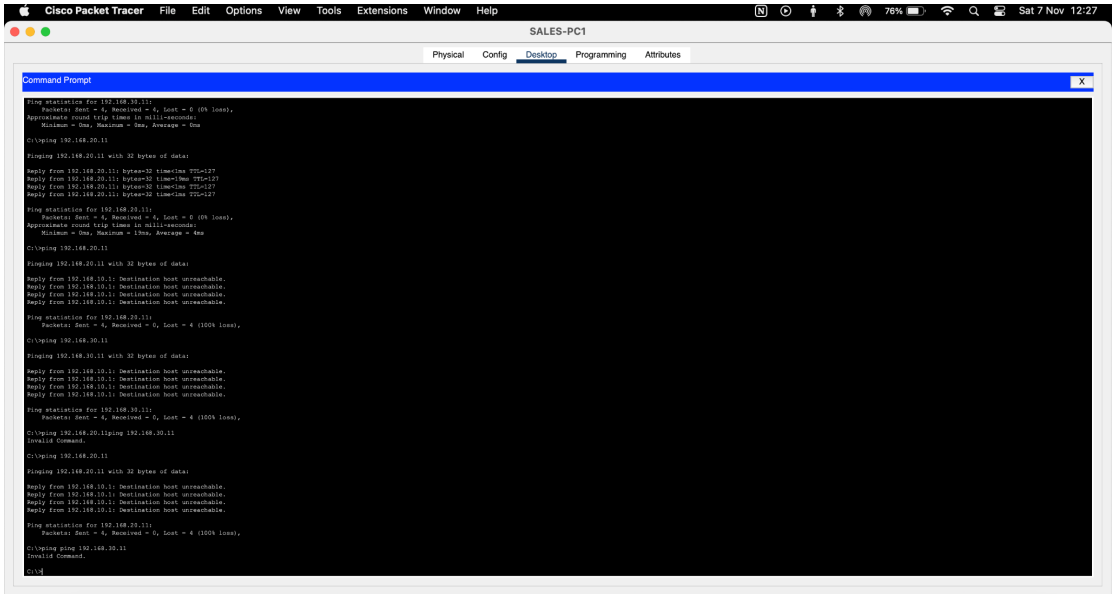


Figure 6. Sales-PC1 final tests: access to IT and HR is blocked.

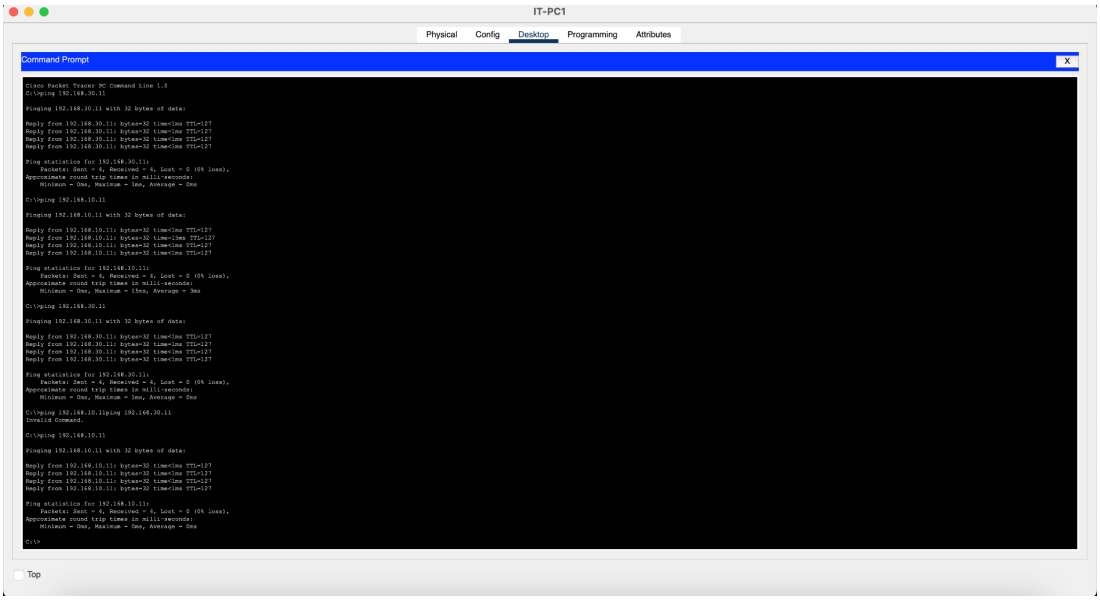


Figure 7. IT-PC1 tests: access to HR and Sales succeeds.

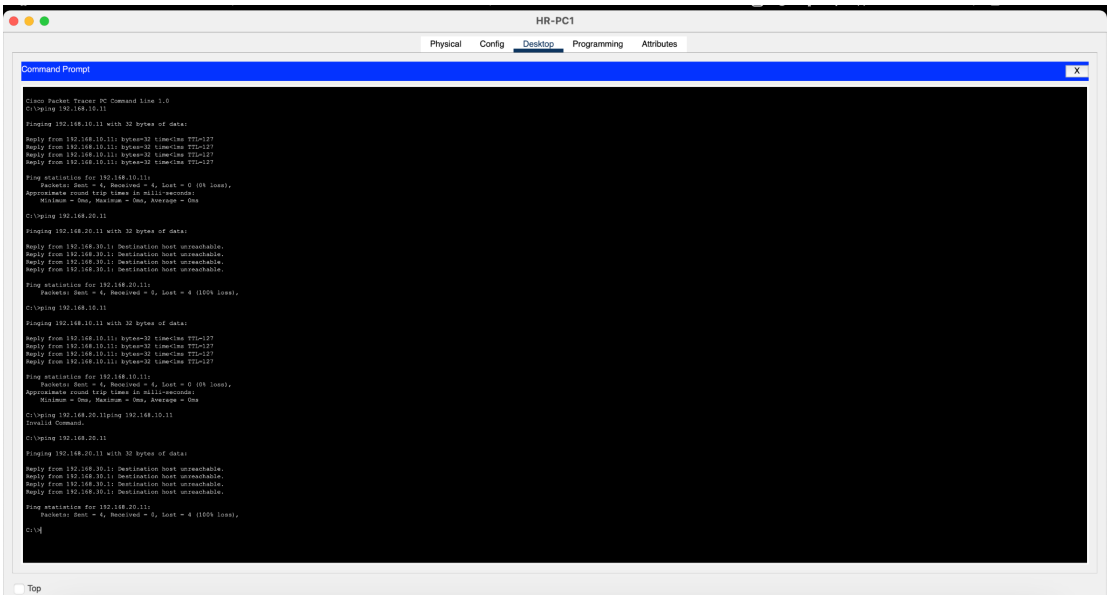


Figure 8. HR-PC1 tests: access to IT is blocked and access to Sales succeeds.

9. Interpretation, limitations and recommended additional validation

The six final ICMP tests match the specified departmental policy. The ACL counters show that configured permit and deny entries have been matched. This is evidence that the rules are active and that the intended ping paths are filtered.

These tests validate ICMP reachability, not every application protocol. The established keyword in a Cisco extended ACL is a TCP flag check, not full stateful firewall inspection. The ICMP echo-reply entries permit replies to ping requests. Other return protocols, including UDP-based application traffic, have not been validated by these screenshots. Test the actual TCP or UDP service required by the scenario before claiming all IT-initiated application sessions work.

A first ping can time out while ARP resolves a destination MAC address. Use repeated test output and packet statistics, rather than a single initial timeout, as evidence.

10. Configuration backup and project files

The router and switch running configurations were copied to startup configuration and both operations returned [OK]. The Packet Tracer title bar showed the project in the November 2026 Networking Projects folder. Recommended project filename: Project 01 - Small Office Network.pkt.

Keep the topology, switch, router, ACL and endpoint test screenshots in a folder named Project 01 - Evidence. Reopen the .pkt file and repeat the verification commands to confirm that the saved state persists.

11. Conclusion

Project 01 implements a small office network with three departmental VLANs, router-on-a-stick inter-VLAN routing, DHCP and extended ACL-based access restrictions. The final endpoint evidence demonstrates that Sales cannot reach IT or HR, IT can reach Sales and HR, and HR cannot reach IT but can reach Sales. Both network device configurations were saved. The project meets the stated ICMP connectivity and access-control objectives, with additional application-layer testing recommended for full protocol coverage.

Appendix A. Useful verification commands

```
! Switch<br/>show vlan brief<br/>show interfaces trunk<br/>show running-config<br/><br/>! Router<br/>show ip interface brief<br/>show ip dhcp binding<br/>show access-lists<br/>show ip interface gigabitEthernet 0/0.10<br/>show ip interface gigabitEthernet 0/0.30<br/>show running-config<br/><br/>! Save on each device<br/>copy running-config startup-config
```

Appendix B. Evidence index

Figure	Evidence item	Screenshot time
1	Logical network topology	12:16:33
2	VLAN membership	12:12:20
3	802.1Q trunk status	12:12:54
4	Router ACL entries and counters	12:21:41
5	ACL attachment to Sales and HR subinterfaces	12:24:03
6	Sales blocked connectivity tests	12:27:00
7	IT successful connectivity tests	12:27:44
8	HR allowed and blocked connectivity tests	12:28:33